

Sami MEZGHANI

Software Engineer

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Education

- 2024–2025 **Exchange Student**, *Karlsruhe University of Applied Sciences*, Germany
- 2022–2025 **Engineering Cycle, Major**, *Higher Institute of Computer Science and Multimedia of Sfax*, Tunisia
- 2020–2022 **Integrated Preparatory, Major**, *Higher Institute of Computer Science and Multimedia of Sfax*, Tunisia

Professional Experience

- March–
August 2025 **Final-Year Project**, *Karlsruhe University of Applied Sciences / Bitzer Academy*, Germany
- Created a 3D model of the Bitzer building and developed an Escape Room game in Unity, applying OOP principles and design patterns (Singleton, Observer, State).
 - Pre-configured and generalized Azure VMs, creating a reusable system image for automated deployment of instances.
 - Developed a REST API (Flask) to automate provisioning and manage the lifecycle of VMs.
 - Set up a reverse proxy to manage communication between the frontend and the streaming environment.
 - Automated VM configuration using Azure Run Command, executing batch scripts to finalize the streaming setup.
- Technologies:** Unity, Blender, C#, Flask, REST API, Azure, NiceDCV, Nginx, OpenSSL, Angular.
- July–
August 2024 **Software Engineering Intern**, *Sofrecom*, Sfax, Tunisia
- Implemented the UCP protocol in Python to communicate with Orange's SMSC.
 - Developed a REST API (Flask) for UCP session management, shortcode validation, and SMS delivery to Orange France numbers.
 - Built a simple Angular frontend to test the API.
 - Set up an SMS simulator to validate UCP message flows before deploying on Orange servers.
 - Analyzed UCP traffic with Wireshark.
- Technologies:** Python, Flask, Angular, UCP, TCP/IP, Git, Wireshark.
- June–
August 2023 **DevOps Intern**, *Spark IT*, Sfax, Tunisia
- Implemented Infrastructure as Code (IaC) using Terraform on Proxmox.
 - Secured shared infrastructure state with GitLab backend and state locking.
 - Designed and optimized GitLab CI/CD pipelines covering the full lifecycle (plan, apply, destroy).
- Technologies:** Terraform, GitLab CI, Proxmox.

Projects

- 2025 **Microservices Architecture for Currency Exchange Application**
- Designed and developed a microservices architecture using Spring Boot and Spring Cloud.
 - Containerized services using Docker and deployed with Docker Compose, then migrated to Kubernetes.
 - Implemented load balancing and auto-scaling with Kubernetes HPA to improve performance.
 - Used Spring Boot Actuator for Kubernetes probes, ensuring high availability and resilient updates.
- Technologies:** Spring Boot, Spring Cloud (Eureka, Config, Gateway), Docker, Kubernetes, REST API, MySQL.
- 2024 **Freelance Project: Web Application for Residence Booking**
- Developed a web application with an admin panel to manage site content and synchronize real-time availability with external platforms.
- Technologies:** REST API, Laravel, Laravel Sanctum, Angular, MySQL, Postman.
- 2024 **CI/CD Pipeline for Spring Boot & Angular Application**
- Developed and implemented a CI/CD pipeline with GitHub Actions to build Docker images, publish them to Docker Hub, and deploy the application to a VM.
- Technologies:** Spring Boot, Angular, MySQL, GitHub Actions, Docker, Docker Compose.

Extracurricular Activities

Competitive Programming

- 2024: Hello World 4.0 – 3rd / 54
- 2023: IEEE Xtreme 17.0 – 31st / 622 (Tunisia), 961st / 4,231 globally
 - Hello World 3.0 – 1st / 45
- 2022: Meta HackerCup – Qualified 1303 / 27,604
 - Round 1: 2956 / 12,330
 - Round 2: 2993 / 5,047
 - IEEE Xtreme 16.0 – 23rd / 448 (Tunisia), 533rd / 2,992 globally
 - Hello World 2.0 – 1st / 23
- Delivered training sessions and CP courses for university students.
- Organized programming contests to prepare students for IEEE Xtreme.

Languages

French • English • Arabic • German: Beginner